



Network Infrastructure Architecture Report

Tech Solutions (SARL)

Version 1.0 — January 8, 2026

Executive Summary

This document provides a complete professional overview of the network infrastructure designed for Tech Solutions (SARL). The architecture simulates a modern enterprise network using GNS3, featuring routed backbone segments, VLAN segmentation, firewall policies, and Linux-based services.

The objective is to model a scalable, secure, and realistic environment for engineering, training, and testing purposes. It provides system engineers and network architects with a clear understanding of the network's logical and physical design, implementation steps, and operational considerations.

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1 Introduction

The Tech Solutions (SARL) network architecture project is a fully simulated enterprise environment developed within GNS3. It includes multi-router infrastructure, departmental VLANs, dynamic routing, and server resources.

The purpose of this report is to present:

- network design principles,
- device roles and interconnections,
- used technologies and protocols,
- deployment and implementation steps,
- configuration and testing methodology.

1.1 Departmental Allocation and Logical Topology

After the integration of the Internet service, the ISP allocated the address space **172.24.0.0/14** to the enterprise. The logical topology and IP allocation for each department are summarized below:

Department	Local Router	Service / VM	PC Clients	Role in the Enterprise
Web / Marketing	RZ-1	Web Server	8905	Manages the website and customer-facing interactions
Supervision / IT	RZ-2	Monitoring Server	1465	Monitors server and network performance
Base de données / Gestion	RZ-3	Database Server	489	Centralizes and secures client and internal data
Partage / Collaboration	RZ-4	NFS Server	165	Enables internal file and document sharing

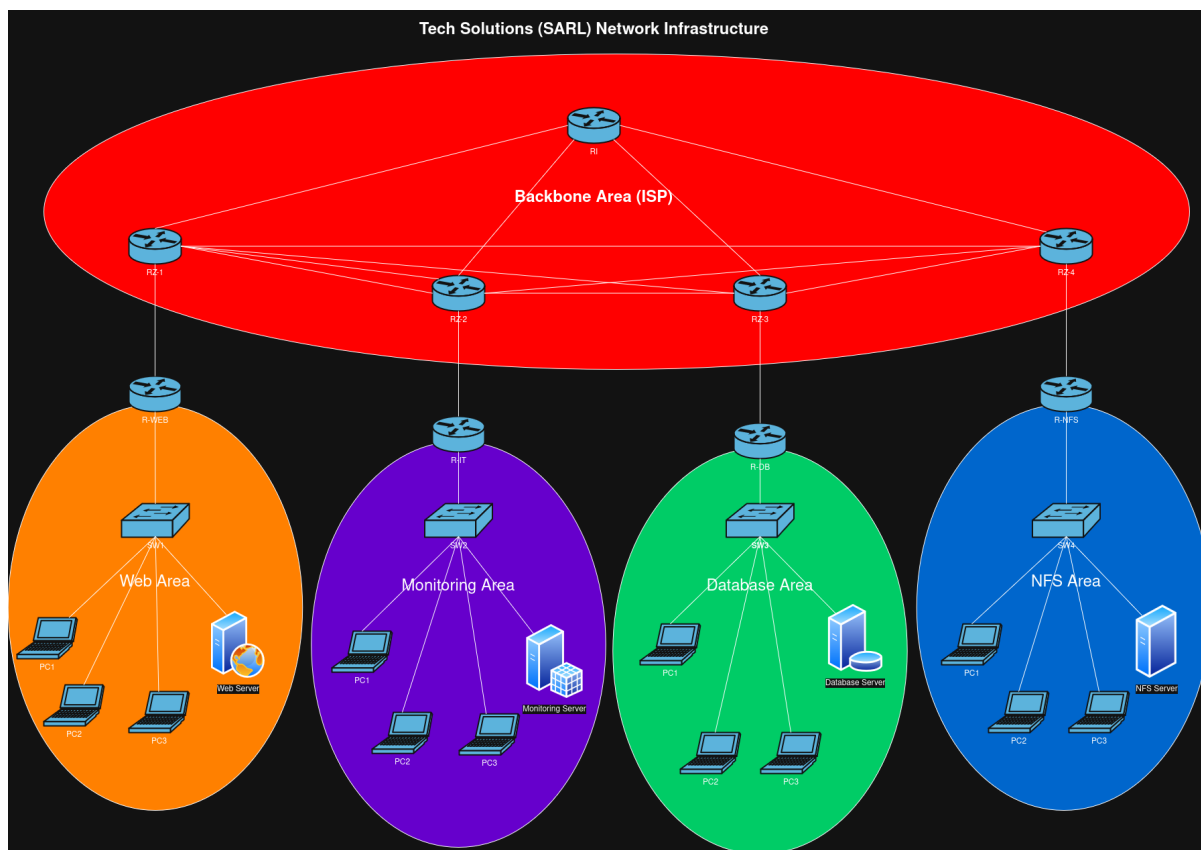
1.2 Subnetting (VLSM) Diagram

The network 172.24.0.0/14 has 262142 hosts. Your subnets need 11032 hosts.								
NAME	HOSTS NEEDED	HOSTS AVAILABLE	UNUSED HOSTS	NETWORK ADDRESS	SLASH	MASK	USABLE RANGE	BROADCAST
Web	8907	16382	7475	172.24.0.0	/18	255.255.192.0	172.24.0.1 - 172.24.63.254	172.24.63.255
IT	1467	2046	579	172.24.64.0	/21	255.255.248.0	172.24.64.1 - 172.24.71.254	172.24.71.255
DB	491	510	19	172.24.72.0	/23	255.255.254.0	172.24.72.1 - 172.24.73.254	172.24.73.255
NFS	167	254	87	172.24.74.0	/24	255.255.255.0	172.24.74.1 - 172.24.74.254	172.24.74.255

2 Technologies Used

- **GNS3 3.0.5** — Network simulation environment.
- **Cisco Routers (3725/3745)** — Core routing infrastructure.
- **Ubuntu Server 20.04** — Server and network service hosting.
- **OSPF** — Dynamic routing protocol.
- **VLANs** — Logical segmentation and broadcast containment.
- **Nmap & Wireshark** — Analysis and testing tools.
- **ZeroTier** — Potential remote access overlay network.
- **SSH / Telnet** — Device management protocols.

3 Architecture Design



3.1 Core Layer

The backbone contains the primary routing equipment enabling communication between all segments.

3.2 Distribution Layer

VLAN segmentation and policy enforcement occur at this layer.

3.3 Access Layer

End-user devices and servers connect at this level.

4 Implementation Steps

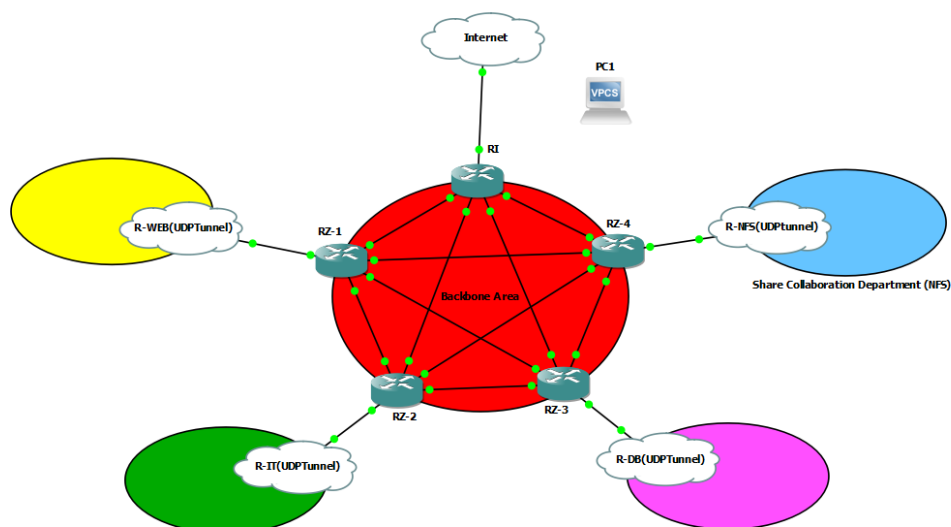
1. Prepare GNS3 environment and import appliance templates.
2. Deploy routers, switches, and servers.
3. Configure VLANs and interfaces.
4. Configure OSPF across backbone routers.
5. Implement ACLs and security policies.
6. Verify connectivity and routing.
7. Test bandwidth, latency, and redundancy.

5 Security Model

Security policies were applied on multiple layers:

- ACLs restricting cross-department traffic.
- VLAN isolation.
- SSH-enabled secure device management.
- Logging and monitoring using Linux servers.

6 Backbone Infrastructure



6.1 Purpose

The backbone provides high-availability routed connectivity between all departments, the Internet, and security services. It ensures redundancy, dynamic routing, and controlled external access.

6.2 Architecture Overview

The backbone consists of:

- Four core routers (CR1–CR4)
- OSPF dynamic routing
- FortiGate firewall (VPN + filtering)
- Internet edge router with NAT

6.3 OSPF Design

- OSPF Area 0 (Backbone)
- All core routers participate in Area 0
- Point-to-point links between core routers
- Route summarization applied toward distribution routers

6.4 VPN and Security

- FortiGate provides site-to-site and remote-access VPN

- Encrypted tunnels secure management and inter-site traffic
- Firewall policies restrict inbound/outbound traffic

6.5 Internet Router (NAT)

- Static default route toward ISP
- PAT (NAT overload) for internal subnets
- Controlled port forwarding for public services

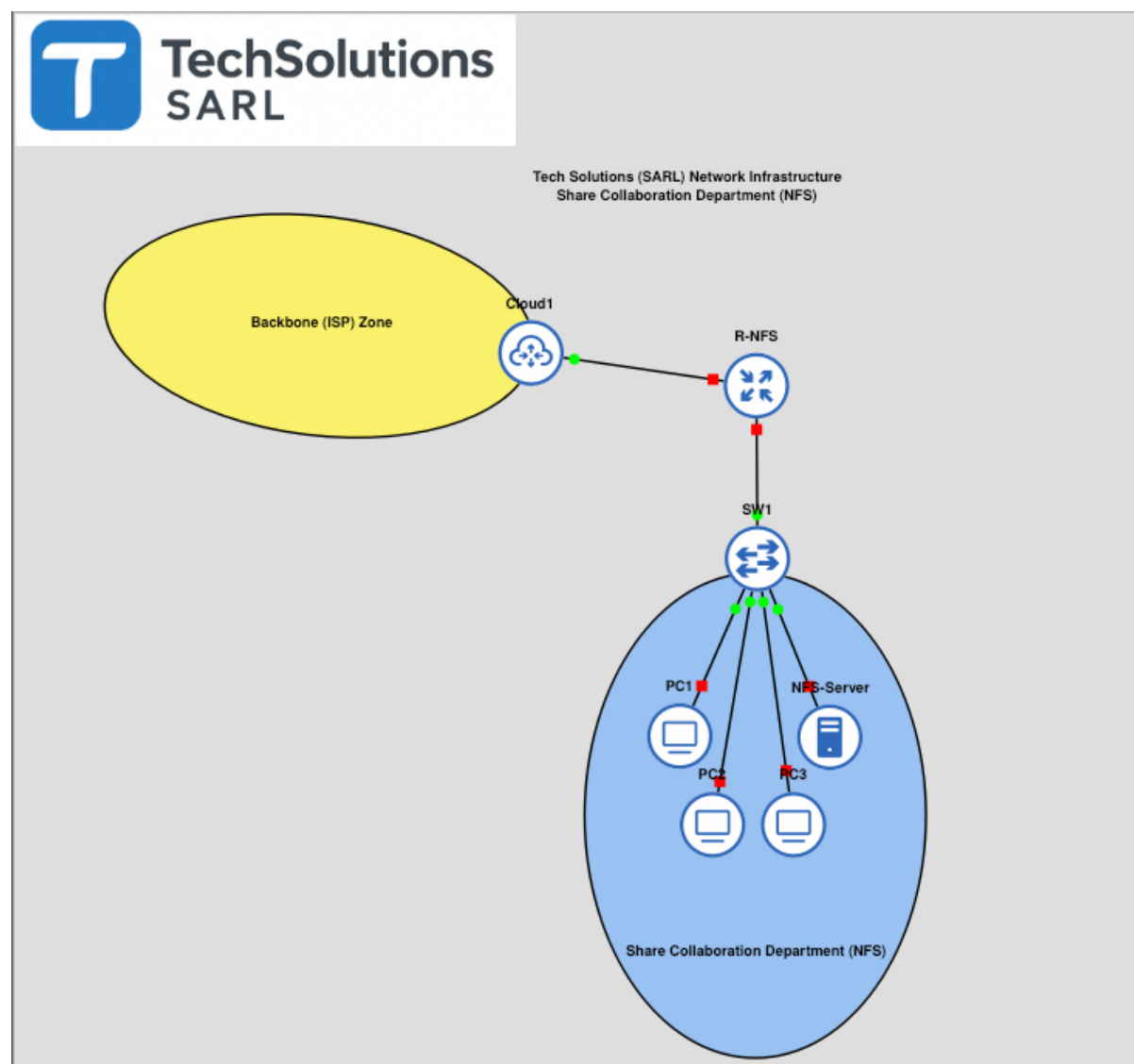
6.6 Screenshots

6.6.1 Backbone - OSPF Protocol

```
RI#show ip route ospf
 200.200.20.0/30 is subnetted, 10 subnets
O    200.200.20.20 [110/11] via 200.200.20.10, 00:03:56, FastEthernet1/0
      [110/11] via 200.200.20.2, 00:03:56, FastEthernet0/0
O    200.200.20.16 [110/11] via 200.200.20.2, 00:03:56, FastEthernet0/0
O    200.200.20.28 [110/2] via 200.200.20.10, 00:03:56, FastEthernet1/0
O    200.200.20.24 [110/11] via 200.200.20.14, 00:03:56, FastEthernet2/0
      [110/11] via 200.200.20.2, 00:03:56, FastEthernet0/0
O    200.200.20.36 [110/2] via 200.200.20.14, 00:03:56, FastEthernet2/0
      [110/2] via 200.200.20.10, 00:03:56, FastEthernet1/0
O    200.200.20.32 [110/2] via 200.200.20.14, 00:03:56, FastEthernet2/0
172.24.0.0/16 is variably subnetted, 3 subnets, 3 masks
O    172.24.0.0/18 [110/30] via 200.200.20.2, 00:03:56, FastEthernet0/0
O    172.24.64.0/21 [110/13] via 200.200.20.14, 00:03:56, FastEthernet2/0
      [110/13] via 200.200.20.10, 00:03:56, FastEthernet1/0
O    172.24.72.0/23 [110/12] via 200.200.20.10, 00:03:58, FastEthernet1/0
10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
O    10.2.0.0/30 [110/3] via 200.200.20.14, 00:03:58, FastEthernet2/0
      [110/3] via 200.200.20.10, 00:03:58, FastEthernet1/0
O    10.3.0.0/30 [110/20] via 200.200.20.2, 00:03:58, FastEthernet0/0
O    10.0.0.0/24 [110/2] via 200.200.20.14, 00:03:58, FastEthernet2/0
O    10.1.0.0/30 [110/2] via 200.200.20.10, 00:03:58, FastEthernet1/0
RI#
```

Figure 1: OSPF Active On Backbone

7 NFS Department



7.1 Purpose

This department provides centralized file sharing services for internal collaboration and data exchange.

7.2 Architecture

- Linux NFS Server (Ubuntu Server)
- Client workstations
- Dedicated router and access switch

7.3 Server Configuration Overview

- NFS service enabled

- Shared directories defined in `/etc/exports`
- Access restricted to internal subnets

7.4 Client Configuration Script Overview

Clients mount shared directories automatically at boot using:

- `mount -t nfs`
- `/etc/fstab`

7.5 Router Script Overview

- Interface IP configuration
- OSPF advertisement
- ACL allowing NFS traffic only

7.6 Switch Script Overview

- VLAN assignment
- Access ports for clients
- Trunk port toward router

7.7 Scripts and Configuration

7.7.1 NFS Router Configuration

```
1  ! ZouariOmar <zouariomar20@gmail.com>
2  ! Since:          2025-11-19
3  ! LastUpdate:     2026-01-05
4  ! RouterName:     R-nfs
5  ! RouterVersion:  C3745
6  ! COPY-THAN-PASTE
7
8  conf t
9  hostname R-nfs
10
11 ! =====
12 ! ===== INTERFACES PART =====
13 ! === FO/1(WAN) - FO/0(LAN) ===
14 ! =====
15
16 ! === FastEthernet0/1 (WAN) ===
17 interface fa0/1
18     description R-nfs to UDP Tunnel
19     ip address 10.0.0.2 255.255.255.0
20     no shutdown
21     exit
22
```

```
23 ! === FastEthernet0/0 (LAN) ===
24 interface fa0/0
25     description R-nfs to SW1
26     ip address 172.24.74.1 255.255.255.0
27     no shutdown
28     exit
29
30 ! =====
31 ! ===== SERVICES PART =====
32 ! ===== OSPF - DHCP - DNS =====
33 ! =====
34
35 ! === OSPF ===
36 router ospf 1
37     router-id 7.7.7.7
38     network 10.0.0.0 0.0.0.255 area 0
39     network 172.24.74.0 0.0.0.255 area 0
40     passive-interface default
41     no passive-interface FastEthernet0/1
42     ! NOTE: 'f0/0' should not be no 'passive-interface'
43     default-information originate
44     exit
45
46 ! Default route toward backbone
47 ip route 0.0.0.0 0.0.0.0 10.0.0.1
48
49 ! === DHCP ===
50 ip dhcp pool NFS-DHCP
51     network 172.24.74.0 255.255.255.0
52     default-router 172.24.74.1
53     dns-server 8.8.8.8
54     exit
55
56 ! Exclude @GW
57 ip dhcp excluded-address 172.24.74.1
58
59 ! Exclude @NFS-server
60 ip dhcp excluded-address 172.24.74.2
61
62 ! === DNS ===
63 ip domain-lookup
64     ip name-server 8.8.8.8
65
66 ! =====
67 ! ===== SECURITY PART =====
68 ! == CONSOLE - SSH - TELNET - ACL ==
69 ! =====
70
71 ! Create an Admin user
72 username admin privilege 15 secret <STRONG_PASSWORD>
73
```

```
74 ! Encrypt all passwords
75 service password-encryption
76
77 ! === Console ===
78 line console 0
79     login local
80     exec-timeout 5 0      ! Auto-logout after 5min
81     logging synchronous ! Clean console display
82     exit
83
84 ! === SSH ===
85 crypto key generate rsa usage-keys label SSH modulus 2048
86 ip ssh version 2
87 ! ip access-list standard SSH_ONLY
88 !     permit X.X.X.X X.X.X.X
89 !     deny any
90 line vty 0 4
91     login local
92     transport input ssh
93 ! access-class SSH_ONLY in
94     exec-timeout 5 0
95     logging synchronous
96     exit
97
98 ! === Telnet ===
99 ! ip access-list standard TELNET_ONLY
100 !     permit 10.0.0.0 0.0.0.255
101 !     deny any
102 line vty 5 15
103     login local
104     transport input telnet
105 ! access-class TELNET_ONLY in
106     exec-timeout 5 0
107     logging synchronous
108     end
109
110 ! Disable unused services
111 ! no ip http server          ! HTTP
112 ! no ip http secure-server  ! HTTPS
113 ! no cdp run                 ! CDP
114
115 write
116
117 ! =====
118 ! === VERIFICATION PART ===
119 ! =====
120
121 ! === OSPF Interfaces ===
122 ! Check which interfaces run OSPF and their state
123 show ip ospf interface brief
124 show ip ospf interface
```



```
125 show ip protocols
126
127 ! === OSPF Interfaces ===
128 ! Verify adjacency with other routers (if backbone is up)
129 show ip ospf neighbor
130 show ip ospf neighbor detail
131
132 ! === OSPF Routes ===
133 ! Verify OSPF learned routes
134 show ip route ospf
135 show ip route
136
137 ! === Default Route / Redistribution ===
138 ! Check default route advertisement
139 show ip route 0.0.0.0
140 show running-config | include default-information
141
142 ! === LAN Connectivity / DHCP ===
143 ! Verify DHCP bindings and pool
144 show ip dhcp binding
145 show ip dhcp pool
146
147 ! === DNS Resolution ===
148 ! Verify DNS lookup from router
149 ping 8.8.8.8
150 ping www.google.com
```

Listing 1: NFS Router Configuration

7.7.2 NFS Server Script

```
1 #!/usr/bin/env bash
2 # =====
3 # .FILE
4 #     nfs-server-conf
5 #
6 # .SYNOPSIS
7 #     Bash script to configure NFS-Linux server
8 #
9 # .DESCRIPTION
10 #     I write this script for
11 #     'Linux nfs-server 6.8.0-88-generic #89-Ubuntu SMP
12 #     PREEMPT_DYNAMIC Sat Oct 11 01:02:46 UTC 2025 x86_64 x86_64 GNU
13 #     /Linux'
14 #     (ubuntu-24.04.3-live-server-amd64)
15 #
16 # .PARAMETER NONE
17 #     NONE
18 #
19 # .INPUTS
20 #     NONE
```

```

19 #
20 # .OUTPUTS
21 #   NONE
22 #
23 # .NOTES
24 #   Version      : 1.0
25 #   Author       : @ZouariOmar <zouariomar20@gmail.com>
26 #   Created      : 11/22/2025
27 #   Change Log   : 9780b36
28 #
29 # .EXAMPLE
30 #   > ./nfs-server-conf
31 #   --
32 #   > NET_INTERFACE="enp2s0" \
33 #     NET_ADDRESS="172.24.74.2/24" \
34 #     NET_GW="172.24.74.1" \
35 #     NET_DNS_SERVERS="[8.8.8.8]" \
36 #     SHARED_DIR="/srv/nfs/share" \
37 #     NFS_GROUP='zouari_omar'
38 #     ./nfs_server_conf
39 # =====
40
41 #####
42 ### CUSTOM CONFIGURATION PART ###
43 #####
44 NET_INTERFACE="enp2s0"
45 NET_SUBNET='X.X.X.X/XX'
46 NET_ADDRESS="X.X.X.X/XX"
47 NET_GW="X.X.X.X"
48 NET_DNS_SERVERS="[X.X.X.X]"
49 SHARED_DIR='path_to_shared_directory'
50 NFS_GROUP='group_to_chown_shared_dir'
51
52 #####
53 ### GENERAL CONFIGURATION PART ###
54 #####
55
56 # Set keymap into fr
57 sudo sed -i 's/XKBLAYOUT=".*"/XKBLAYOUT="fr"/' /etc/default/
    keyboard
58
59 # Update the system
60 sudo apt update -y && sudo apt upgrade -y
61
62 # Change the current hostname
63 hostnamectl set-hostname nfs-server
64
65 #####
66 ### NETWORK CONFIGURATION PART ###
67 #####
68

```

```
69 # Set the private static ip address
70 sudo bash -c "cat << 'EOF' > /etc/netplan/01-static.yaml
71 network:
72   version: 2
73   renderer: networkd
74   ethernets:
75     ${NET_INTERFACE}:
76       dhcp4: no
77       addresses:
78         - ${NET_ADDRESS}
79       nameservers:
80         addresses: ${NET_DNS_SERVERS}
81       routes:
82         - to: default
83           via: ${NET_GW}
84 EOF"
85
86 # Change 01-static.yaml accessibility
87 sudo chmod 600 /etc/netplan/01-static.yaml
88
89 # Apply changes
90 sudo netplan apply
91
92 #####
93 ### NFS-SERVER CONFIGURATION PART ###
94 #####
95
96 # Install Server NFS-SERVER package
97 sudo apt install -y nfs-kernel-server nfs-common
98
99 # Enable NFS service
100 sudo systemctl enable --now nfs-server
101
102 # Create and Export a Shared Directory
103 sudo mkdir -p ${SHARED_DIR}
104
105 # Add local subnet to the ACL (Access Control List)
106 echo "${SHARED_DIR} ${NET_SUBNET}(rw, sync, no_subtree_check)" |
    sudo tee -a /etc/exports
107
108 # Apply exports
109 sudo exportfs -ra
110
111 # Give grp permission
112 # NOTE: To add new group 'sudo groupadd NFS_GROUP'
113 # NOTE: To delete group 'sudo groupdel NFS_GROUP'
114 sudo chown -R :${NFS_GROUP} ${SHARED_DIR}
115 sudo chmod -R 775 ${SHARED_DIR}
116
117 # Configure Firewall
118 sudo ufw allow from ${NET_SUBNET} to any port nfs
```

Listing 2: NFS Server Configuration Script

7.7.3 NFS Client Script

```

1  #!/usr/bin/env bash
2  # =====
3  # .FILE
4  #   nfs_client_conf
5  #
6  # .SYNOPSIS
7  #   Bash script to configure NFS-Linux client
8  #
9  # .DESCRIPTION
10 #   I write this script for
11 #   'Linux nfs-server 6.8.0-88-generic #89-Ubuntu SMP
    PREEMPT_DYNAMIC Sat Oct 11 01:02:46 UTC 2025 x86_64 x86_64 GNU
    /Linux '
12 #   (ubuntu-24.04.3-live-server-amd64)
13 #
14 # .PARAMETER NONE
15 #   NONE
16 #
17 # .INPUTS
18 #   NONE
19 #
20 # .OUTPUTS
21 #   NONE
22 #
23 # .NOTES
24 #   Version      : 1.0
25 #   Author       : @ZouariOmar <zouariomar20@gmail.com>
26 #   Created      : 11/22/2025
27 #   Change Log   : 9780b36
28 #
29 # .EXAMPLE
30 #   > ./nfs-client-conf
31 #   --
32 #   > NET_INTERFACE=enp2s0          \
33 #     SERVER_ADDRESS='172.24.74.2' \
34 #     SHARED_DIR='/srv/nfs/share'   \
35 #     MOUNT_POINT='/mnt/nfs'
36 #     ./nfs_client_conf
37 #   =====
38
39 #####
40 ### CUSTOM CONFIGURATION PART ###
41 #####
42 NET_INTERFACE="enp2s0"
43 SERVER_ADDRESS="X.X.X.X"
44 SHARED_DIR='path_to_shared_directory'

```

```
45 MOUNT_POINT='path_to_mount_directory'
46
47 #####
48 ### GENERAL CONFIGURATION PART ###
49 #####
50
51 # Set keymap into fr
52 sudo sed -i 's/XKBLAYOUT=".*"/XKBLAYOUT="fr"/' /etc/default/
    keyboard
53
54 # Update the system
55 sudo apt update -y && sudo apt upgrade -y
56
57 # Change the current hostname
58 hostnamectl set-hostname nfs-client
59
60 #####
61 ### NETWORK CONFIGURATION PART ###
62 #####
63
64 # Set dhcp ip address
65 sudo bash -c "cat << 'EOF' > /etc/netplan/01-enp2s0.yaml
66 network:
67   version: 2
68   ethernets:
69     ${NET_INTERFACE}:
70       dhcp4: true
71 EOF"
72
73 # Change 01-enp2s0.yaml accessibility
74 sudo chmod 600 /etc/netplan/01-enp2s0.yaml
75
76 # Apply changes
77 sudo netplan apply
78
79 #####
80 ### NFS-CLIENT CONFIGURATION PART ###
81 #####
82
83 # INFO: To change group name: 'sudo groupmod -n new-name current-
    name'
84 # Install NFS-CLIENT package
85 sudo apt install nfs-common
86
87 # Create mount point
88 sudo mkdir -p ${MOUNT_POINT}
89
90 # Mount the share
91 sudo mount ${SERVER_ADDRESS}:${SHARED_DIR} ${MOUNT_POINT}
92
93 # Verify mount
```

```

94 df -h | grep nfs
95
96 # Umount the share
97 # sudo umount ${MOUNT_POINT}
98
99 # Mount NFS share safely using systemd automount; system boots
   normally even if the NFS server is down.
100 echo "${SERVER_ADDRESS}:${SHARED_DIR} ${MOUNT_POINT} nfs
   defaults,_netdev,x-systemd.automount,nofail 0 0" | sudo tee
   -a /etc/fstab >/dev/null

```

Listing 3: NFS Client Configuration Script

7.8 Screenshots

7.8.1 NFS Test

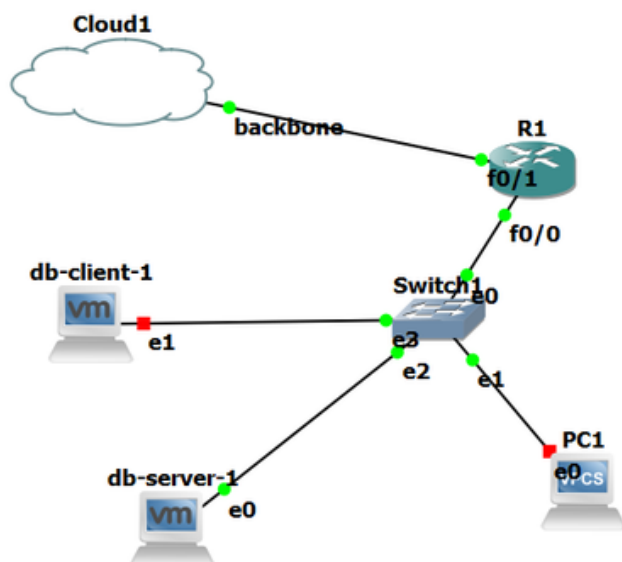
```

zouari_omar@nfs-server:~$ cat /srv/nfs/share/*
Hello Esprit!
Helloooo
zouari_omar@nfs-server:~$ _

```

Figure 2: NFS Test On ‘/srv/nfs/share’

8 Database Department



8.1 Purpose

This department hosts the enterprise MySQL database, centralizing business and client data securely.

8.2 Architecture

- Linux Database Server
- MySQL service
- Restricted client access

8.3 MySQL Configuration

- MySQL bound to internal interface only
- Dedicated database users per service
- Regular backups scheduled

8.4 Security

- ACLs limiting database access
- Strong authentication
- Network isolation via VLAN

8.5 Scripts and Configuration

8.5.1 Database Router Configuration

```
1  ! ZouariOmar <zouariomar20@gmail.com>
2  ! Since:          2025-12-23
3  ! LastUpdate:     2026-01-05
4  ! RouterName:     R-db
5  ! RouterVersion:  C3745
6  ! COPY-THAN-PASTE
7
8  conf t
9  hostname R-db
10
11 ! =====
12 ! ===== INTERFACES PART =====
13 ! === F0/1(WAN) - F0/0(LAN) ===
14 ! =====
15
16 ! === FastEthernet0/1 (WAN) ===
17 interface FastEthernet0/1
18     description R-db to UDP Tunnel / Backbone
19     ip address 10.1.0.2 255.255.255.252
20     no shutdown
21     exit
```

```
22
23 ! === FastEthernet0/0 (LAN) ===
24 interface FastEthernet0/0
25     description R-db to DB-LAN
26     ip address 172.24.72.1 255.255.254.0
27     no shutdown
28     exit
29
30 ! =====
31 ! ===== SERVICES PART =====
32 ! ===== OSPF - DHCP - DNS =====
33 ! =====
34
35 ! === OSPF ===
36 router ospf 1
37     router-id 6.6.6.6
38     log-adjacency-changes
39     network 10.1.0.0 0.0.0.3 area 0
40     network 172.24.72.0 0.0.1.255 area 0
41     passive-interface default
42     no passive-interface FastEthernet0/1
43     default-information originate
44     exit
45
46 ! Default route toward backbone
47 ip route 0.0.0.0 0.0.0.0 10.1.0.1
48
49 ! === DHCP ===
50 ! Excluded addresses
51 ip dhcp excluded-address 172.24.72.1 172.24.72.2
52
53 ip dhcp pool DB-LAN
54     network 172.24.72.0 255.255.254.0
55     default-router 172.24.72.1
56     dns-server 8.8.8.8
57     exit
58
59 ! === DNS ===
60 ip domain-lookup
61 ip name-server 8.8.8.8
62 ip name-server 1.1.1.1
63
64 ! =====
65 ! ===== SECURITY PART =====
66 ! CONSOLE - SSH - TELNET - ACL
67 ! =====
68
69 ! Create Admin user
70 username admin privilege 15 secret <STRONG_PASSWORD>
71
72 ! Encrypt all passwords
```



```
73 service password-encryption
74
75 ! === Console ===
76 line console 0
77     login local
78     exec-timeout 5 0
79     logging synchronous
80     exit
81
82 ! === SSH ===
83 crypto key generate rsa usage-keys label SSH modulus 2048
84 ip ssh version 2
85
86 line vty 0 4
87     login local
88     transport input ssh
89     exec-timeout 5 0
90     logging synchronous
91     exit
92
93 ! === Telnet (optional / legacy) ===
94 line vty 5 15
95     login local
96     transport input telnet
97     exec-timeout 5 0
98     logging synchronous
99     end
100
101 ! Disable unused services
102 no ip http server
103 no ip http secure-server
104 ! no cdp run
105
106 write
107
108 ! =====
109 ! === VERIFICATION PART ===
110 ! =====
111
112 ! === OSPF Interfaces ===
113 ! Check which interfaces run OSPF and their state
114 show ip ospf interface brief
115 show ip ospf interface
116 show ip protocols
117
118 ! === OSPF Interfaces ===
119 ! Verify adjacency with other routers (if backbone is up)
120 show ip ospf neighbor
121 show ip ospf neighbor detail
122
123 ! === OSPF Routes ===
```

```
124 ! Verify OSPF learned routes
125 show ip route ospf
126 show ip route
127
128 ! === Default Route / Redistribution ===
129 ! Check default route advertisement
130 show ip route 0.0.0.0
131 show running-config | include default-information
132
133 ! === LAN Connectivity / DHCP ===
134 ! Verify DHCP bindings and pool
135 show ip dhcp binding
136 show ip dhcp pool
137
138 ! === DNS Resolution ===
139 ! Verify DNS lookup from router
140 ping 8.8.8.8
141 ping www.google.com
```

Listing 4: DB Router Configuration

8.5.2 Database Server Script

```
1 #!/usr/bin/env bash
2 # =====
3 # .FILE
4 #   db_server_conf
5 #
6 # .SYNOPSIS
7 #   Bash script to configure DB-Linux server
8 #
9 # .DESCRIPTION
10 #   I write this script for
11 #   'Linux nfs-server 6.8.0-88-generic #89-Ubuntu SMP
12 #   PREEMPT_DYNAMIC Sat Oct 11 01:02:46 UTC 2025 x86_64 x86_64 GNU
13 #   /Linux'
14 #   (ubuntu-24.04.3-live-server-amd64)
15 #
16 # .PARAMETER NONE
17 #   NONE
18 #
19 # .INPUTS
20 #   NONE
21 #
22 # .OUTPUTS
23 #   NONE
24 #
25 # .NOTES
26 #   Version      : 1.0
27 #   Author       : @ZouariOmar <zouariomar20@gmail.com>
28 #   Created      : 11/22/2025
```

```

27 # Change Log : 9780b36
28 #
29 # .EXAMPLE
30 # > ./db_server_conf
31 # --
32 # > NET_INTERFACE="enp2s0" \
33 # NET_ADDRESS="172.24.74.2/24" \
34 # NET_GW="172.24.74.1" \
35 # NET_DNS_SERVERS="[8.8.8.8]" \
36 # DB_ACCESS_CLIENT="clientdb '@'192.168.2.%'"
37 # ./db_server_conf
38 # =====
39
40 #####
41 ### CUSTOM CONFIGURATION PART ###
42 #####
43 NET_INTERFACE="enp2s0"
44 NET_ADDRESS="X.X.X.X/XX"
45 NET_GW="X.X.X.X"
46 NET_DNS_SERVERS="[X.X.X.X]"
47 DB_ACCESS_CLIENT="'db_client '@'X.X.X.X'"
48
49 #####
50 ### GENERAL CONFIGURATION PART ###
51 #####
52
53 # Set keymap into fr
54 sudo sed -i 's/XKBLAYOUT=".*"/XKBLAYOUT="fr"/' /etc/default/
    keyboard
55
56 # Update the system
57 sudo apt update -y && sudo apt upgrade -y
58
59 # Change the current hostname
60 hostnamectl set-hostname nfs-server
61
62 #####
63 ### NETWORK CONFIGURATION PART ###
64 #####
65
66 # Set the private static ip address
67 sudo bash -c "cat << 'EOF' > /etc/netplan/01-static.yaml
68 network:
69   version: 2
70   renderer: networkd
71   ethernets:
72     ${NET_INTERFACE}:
73       dhcp4: no
74       addresses:
75         - ${NET_ADDRESS}
76       nameservers:

```

```

77     addresses: ${NET_DNS_SERVERS}
78     routes:
79         - to: default
80           via: ${NET_GW}
81 EOF"
82
83 # Change 01-static.yaml accessibility
84 sudo chmod 600 /etc/netplan/01-static.yaml
85
86 # Apply changes
87 sudo netplan apply
88
89 #####
90 ### DB-SERVER CONFIGURATION PART ###
91 #####
92
93 # Install MySQL Server
94 sudo apt install mysql-server -y
95
96 # Enable mysql service
97 sudo systemctl enable --now mysql
98
99 # Allow Remote Connections
100 sudo sed -i 's/^bind-address\s*=\s*/bind-address = 0.0.0.0/' /etc/
    mysql/mysql.conf.d/mysqld.cnf
101 sudo systemctl restart mysql
102
103 # Firewall and Port Check
104 sudo ufw allow 3306/tcp
105 sudo ufw reload
106
107 # Database and User Creation
108 sudo mysql -e "CREATE DATABASE IF NOT EXISTS labdb; CREATE USER
    IF NOT EXISTS ${DB_ACCESS_CLIENT} IDENTIFIED BY 'StrongPass123
    !'; GRANT ALL PRIVILEGES ON labdb.* TO ${DB_ACCESS_CLIENT};
    FLUSH PRIVILEGES;"
109
110 # Create Table and Insert Data
111 sudo mysql -e "USE labdb; CREATE TABLE IF NOT EXISTS users (id
    INT AUTO_INCREMENT PRIMARY KEY, username VARCHAR(50), email
    VARCHAR(100)); INSERT INTO users (username, email) VALUES ('
    server_user1','srv1@mail.com'),('server_user2','srv2@mail.com
    '); SELECT * FROM users;"

```

Listing 5: DB Server Configuration Script

8.5.3 Database Client Script

```

1 #!/usr/bin/env bash
2 # =====
3 # .FILE

```

```

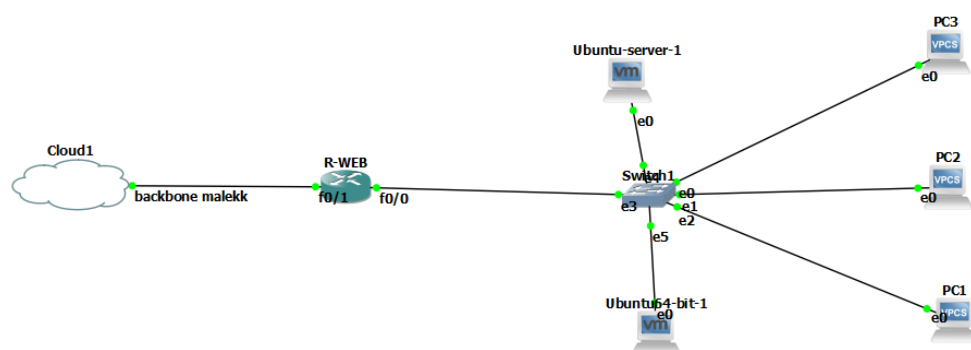
4 # db_client_conf
5 #
6 # .SYNOPSIS
7 #   Bash script to configure DB-Linux client
8 #
9 # .DESCRIPTION
10 #   I write this script for
11 #   'Linux nfs-server 6.8.0-88-generic #89-Ubuntu SMP
12 #   PREEMPT_DYNAMIC Sat Oct 11 01:02:46 UTC 2025 x86_64 x86_64 GNU
13 #   /Linux'
14 #   (ubuntu-24.04.3-live-server-amd64)
15 #
16 # .PARAMETER NONE
17 #   NONE
18 #
19 # .INPUTS
20 #   NONE
21 #
22 # .OUTPUTS
23 #   NONE
24 #
25 # .NOTES
26 #   Version      : 1.0
27 #   Author       : @ZouariOmar <zouariomar20@gmail.com>
28 #   Created      : 11/22/2025
29 #   Change Log   : 9780b36
30 #
31 # .EXAMPLE
32 #   > ./nfs_client_conf
33 #   --
34 #   > NET_INTERFACE=enp2s0 \
35 #       DB_SERVER_IP_ADDRESS="172.24.72.2" \
36 #       DB_CLIENT_NAME="client" \
37 #       DB_CLIENT_PASSWORD="client123"
38 #       ./db_client_conf
39 #   =====
40 #####
41 ### CUSTOM CONFIGURATION PART ###
42 #####
43 NET_INTERFACE="enp2s0"
44 DB_SERVER_IP_ADDRESS="X.X.X.X"
45 DB_CLIENT_NAME="client_name"
46 DB_CLIENT_PASSWORD="client_password"
47 HOSTNAME="db-client"
48 #####
49 ### GENERAL CONFIGURATION PART ###
50 #####
51
52 # Set keymap into fr

```

```
53 sudo sed -i 's/XKBLAYOUT=".*"/XKBLAYOUT="fr"/' /etc/default/
    keyboard
54
55 # Update the system
56 sudo apt update -y && sudo apt upgrade -y
57
58 # Change the current hostname
59 hostnamectl set-hostname ${HOSTNAME}
60
61 #####
62 ### NETWORK CONFIGURATION PART ###
63 #####
64
65 # Set dhcp ip address
66 sudo bash -c "cat << 'EOF' > /etc/netplan/01-enp2s0.yaml
67 network:
68   version: 2
69   ethernets:
70     ${NET_INTERFACE}:
71       dhcp4: true
72 EOF"
73
74 # Change 01-enp2s0.yaml accessibility
75 sudo chmod 600 /etc/netplan/01-enp2s0.yaml
76
77 # Apply changes
78 sudo netplan apply
79
80 #####
81 ### DB-CLIENT CONFIGURATION PART ###
82 #####
83
84 # Install MySQL Client
85 sudo apt install mysql-client -y
86
87 # Remote Connection && Retrieve Server Data
88 mysql -h ${DB_SERVER_IP_ADDRESS} -u ${DB_CLIENT_NAME} -p -e "SHOW
    DATABASES; USE labdb; SHOW TABLES; SELECT * FROM users;"
```

Listing 6: DB Client Configuration Script

9 Web Department



9.1 Purpose

Hosts the public-facing enterprise website and internal web services.

9.2 Architecture

- Linux Web Server
- Apache2 HTTP server
- Client workstations

9.3 Apache2 Configuration

- Virtual hosts enabled
- Static and dynamic content support
- Logs stored for auditing

9.4 Router and Security

- Port forwarding via NAT
- ACLs permitting HTTP/HTTPS
- OSPF routing integration

9.5 Scripts and Configuration

9.5.1 Web Router Configuration

9.5.2 Web Server Script

```

1 #!/usr/bin/env bash
2 # =====
3 # .FILE

```

```

4 # web_server_conf
5 #
6 # .SYNOPSIS
7 # Bash script to configure WEB-Linux server
8 #
9 # .DESCRIPTION
10 # I write this script for
11 # 'Linux nfs-server 6.8.0-88-generic #89-Ubuntu SMP
    PREEMPT_DYNAMIC Sat Oct 11 01:02:46 UTC 2025 x86_64 x86_64 GNU
    /Linux'
12 # (ubuntu-24.04.3-live-server-amd64)
13 #
14 # .PARAMETER NONE
15 # NONE
16 #
17 # .INPUTS
18 # NONE
19 #
20 # .OUTPUTS
21 # NONE
22 #
23 # .NOTES
24 # Version : 1.0
25 # Author : @ZouariOmar <zouariomar20@gmail.com>
26 # Created : 11/22/2025
27 # Change Log : 9780b36
28 #
29 # .EXAMPLE
30 # > ./web_server_conf
31 # --
32 # > NET_INTERFACE="enp2s0" \
33 # NET_ADDRESS="172.24.74.2/24" \
34 # NET_GW="172.24.74.1" \
35 # NET_DNS_SERVERS="[8.8.8.8]" \
36 # WEB_SERVER_ADDRESS='172.24.74.2'
37 # ./web_server_conf
38 # =====
39
40 #####
41 ### CUSTOM CONFIGURATION PART ###
42 #####
43 NET_INTERFACE="enp2s0"
44 NET_ADDRESS="X.X.X.X/XX"
45 NET_GW="X.X.X.X"
46 NET_DNS_SERVERS="[X.X.X.X]"
47
48 #####
49 ### GENERAL CONFIGURATION PART ###
50 #####
51
52 # Set keymap into fr

```



```

53 sudo sed -i 's/XKBLAYOUT=".*"/XKBLAYOUT="fr"/' /etc/default/
    keyboard
54
55 # Update the system
56 sudo apt update -y && sudo apt upgrade -y
57
58 # Change the current hostname
59 hostnamectl set-hostname web-server
60
61 #####
62 ### NETWORK CONFIGURATION PART ###
63 #####
64
65 # Set the private static ip address
66 sudo bash -c "cat << 'EOF' > /etc/netplan/01-static.yaml
67 network:
68   version: 2
69   renderer: networkd
70   ethernets:
71     ${NET_INTERFACE}:
72       dhcp4: no
73       addresses:
74         - ${NET_ADDRESS}
75       nameservers:
76         addresses: ${NET_DNS_SERVERS}
77       routes:
78         - to: default
79           via: ${NET_GW}
80 EOF"
81
82 # Change 01-static.yaml accessibility
83 sudo chmod 600 /etc/netplan/01-static.yaml
84
85 # Apply changes
86 sudo netplan apply
87
88 #####
89 ### WEB-SERVER CONFIGURATION PART ###
90 #####
91
92 # Install apache2 package
93 sudo apt install apache2 -y
94
95 # Enable apache2 service
96 sudo systemctl enable --now apache2

```

Listing 7: Web Server Configuration Script

9.5.3 Web Client Script

```

1 #!/usr/bin/env bash

```

```

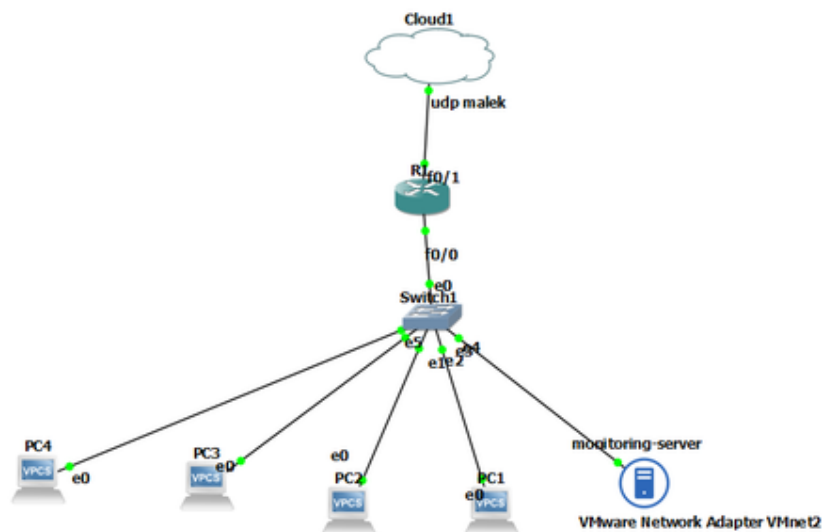
2 # =====
3 # .FILE
4 #   web_client_conf
5 #
6 # .SYNOPSIS
7 #   Bash script to configure WEB-Linux client
8 #
9 # .DESCRIPTION
10 #   I write this script for
11 #   'Linux nfs-server 6.8.0-88-generic #89-Ubuntu SMP
12 #   PREEMPT_DYNAMIC Sat Oct 11 01:02:46 UTC 2025 x86_64 x86_64 GNU
13 #   /Linux '
14 #   (ubuntu-24.04.3-live-server-amd64)
15 #
16 # .PARAMETER NONE
17 #   NONE
18 #
19 # .INPUTS
20 #   NONE
21 #
22 # .OUTPUTS
23 #   NONE
24 #
25 # .NOTES
26 #   Version      : 1.0
27 #   Author       : @ZouariOmar <zouariomar20@gmail.com>
28 #   Created      : 11/22/2025
29 #   Change Log   : 9780b36
30 #
31 # .EXAMPLE
32 #   > ./web_client_conf
33 #   --
34 #   > NET_INTERFACE=enp2s0 \
35 #   WEB_SERVER_ADDRESS='172.24.74.2'
36 #   ./web_client_conf
37 #   =====
38 #####
39 ### CUSTOM CONFIGURATION PART ###
40 #####
41 NET_INTERFACE="enp2s0"
42 WEB_SERVER_ADDRESS="X.X.X.X"
43 #####
44 ### GENERAL CONFIGURATION PART ###
45 #####
46
47 # Set keymap into fr
48 sudo sed -i 's/XKBLAYOUT=".*"/XKBLAYOUT="fr"/' /etc/default/
49 keyboard

```

```
50 # Update the system
51 sudo apt update -y && sudo apt upgrade -y
52
53 # Change the current hostname
54 hostnamectl set-hostname web-client
55
56 #####
57 ### NETWORK CONFIGURATION PART ###
58 #####
59
60 # Set dhcp ip address
61 sudo bash -c "cat << 'EOF' > /etc/netplan/01-enp2s0.yaml
62 network:
63   version: 2
64   ethernets:
65     ${NET_INTERFACE}:
66       dhcp4: true
67 EOF"
68
69 # Change 01-enp2s0.yaml accessibility
70 sudo chmod 600 /etc/netplan/01-enp2s0.yaml
71
72 # Apply changes
73 sudo netplan apply
74
75 #####
76 ### WEB-CLIENT CONFIGURATION PART ###
77 #####
78
79 # Test the Web server
80 curl ${WEB_SERVER_ADDRESS}
```

Listing 8: Web Client Configuration Script

10 Monitoring Department



10.1 Purpose

Provides real-time monitoring, metrics collection, and visualization of the entire infrastructure.

10.2 Architecture

- Linux Monitoring Server
- Prometheus for metrics collection
- Grafana for visualization

10.3 Prometheus

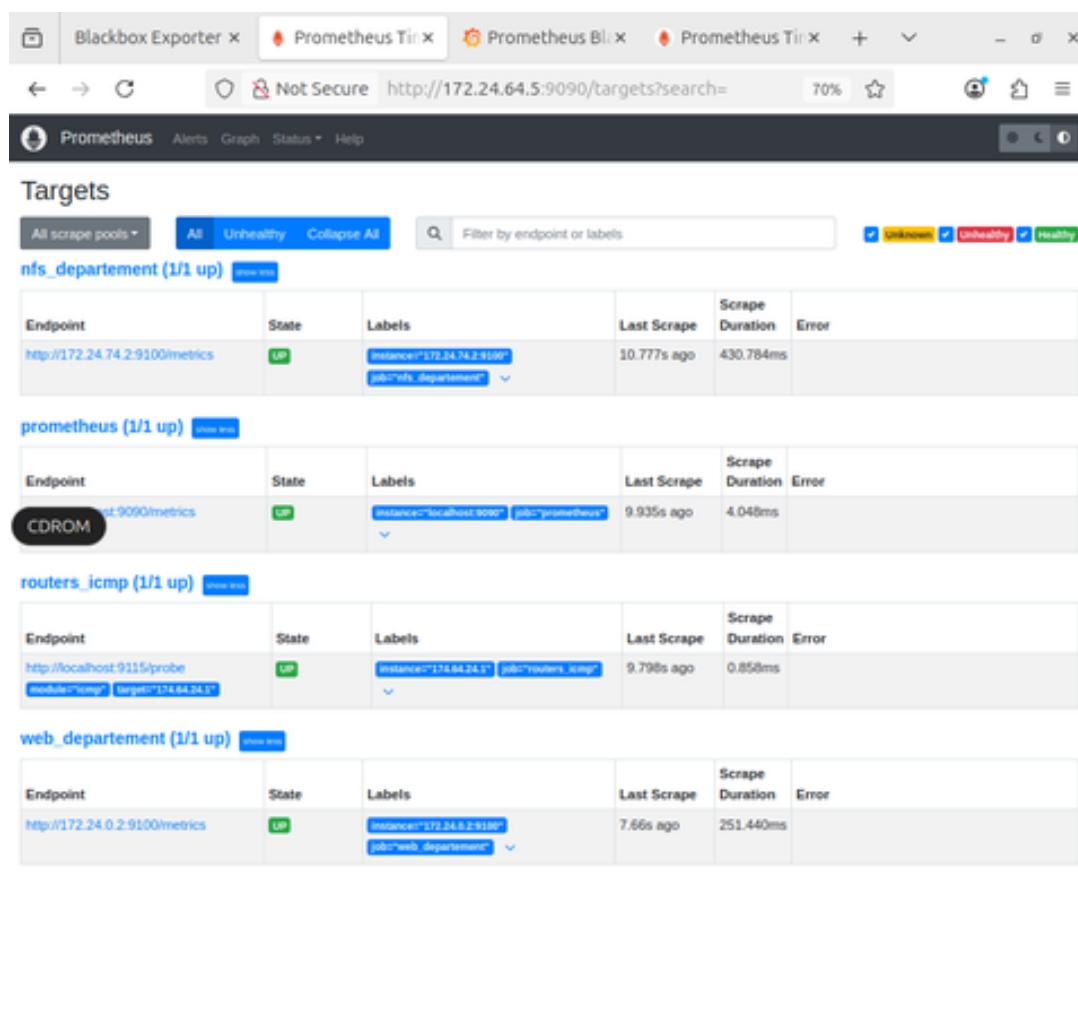
- Scrapes routers and servers
- Node exporter installed on Linux hosts
- Custom alert rules defined

10.4 Grafana

- Dashboards for CPU, RAM, network usage
- Role-based access control
- Historical data analysis

10.5 Screenshots

10.5.1 Prometheus Web Interface



The screenshot displays the Prometheus Web Interface in a web browser. The browser's address bar shows the URL `http://172.24.64.5:9090/targets?search=`. The interface has a dark header with the Prometheus logo and navigation links: Alerts, Graph, Status, and Help. Below the header, the 'Targets' section is active, showing a list of scrape pools. Each pool has a table of targets with columns for Endpoint, State, Labels, Last Scrape, Scrape Duration, and Error. The targets are categorized by scrape pool: nfs_departement, prometheus, routers_icmp, and web_departement. All targets are in a 'UP' state.

Scrape Pool	Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
nfs_departement (1/1 up)	<code>http://172.24.74.2:9100/metrics</code>	UP	<code>instance="172.24.74.2:9100"</code> <code>job="nfs_departement"</code>	10.777s ago	430.784ms	
prometheus (1/1 up)	<code>http://localhost:9090/metrics</code>	UP	<code>instance="localhost:9090"</code> <code>job="prometheus"</code>	9.935s ago	4.048ms	
routers_icmp (1/1 up)	<code>http://localhost:9115/probe</code>	UP	<code>instance="174.64.24.1"</code> <code>job="routers_icmp"</code> <code>module="icmp"</code> <code>target="174.64.24.1"</code>	9.798s ago	0.858ms	
web_departement (1/1 up)	<code>http://172.24.0.2:9100/metrics</code>	UP	<code>instance="172.24.0.2:9100"</code> <code>job="web_departement"</code>	7.66s ago	251.440ms	

Figure 3: Prometheus Web Interface Showing Active Targets

10.5.2 Grafana Dashboard

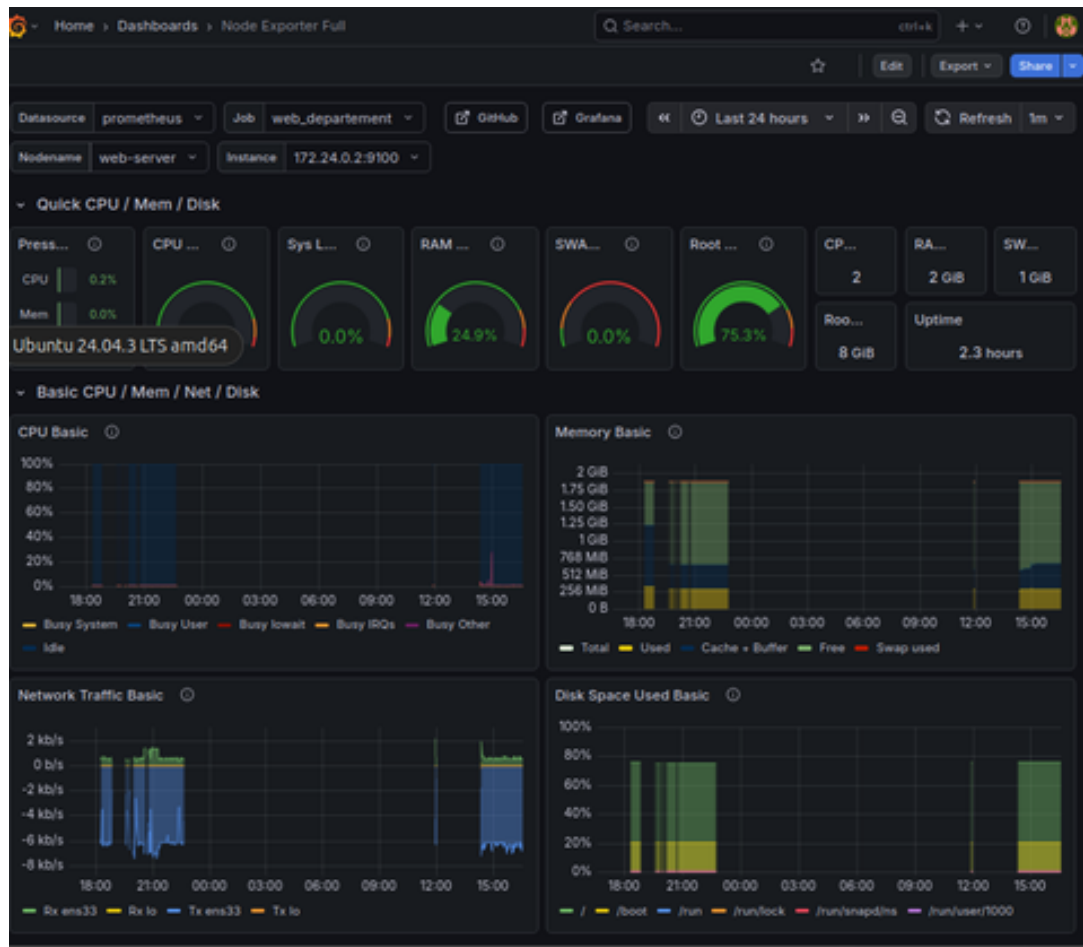


Figure 4: Grafana Node Exporter Dashboard (ID: 1860)

11 Conclusion

This document summarizes the complete network architecture for Tech Solutions (SARL). The simulated environment provides a realistic, scalable, and secure platform for training, experimentation, and enterprise design validation.

End of Report

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